```
C:\Users\270708326\PycharmProjects\PythonProject\SmartHealthCare\.venv\Scripts\python.exe C:\Users\270708326\PycharmProjects\PythonProject\SmartHealthCare\exploratory_data_analysis.py
======================================================================
SMARTCARE EXPLORATORY DATA ANALYSIS
======================================================================

✓ Data loaded: 50 records, 21 variables

======================================================================
1. DEMOGRAPHIC CHARACTERISTICS
======================================================================

Gender Distribution:
  Female: 30 (60.0%)
  Male: 20 (40.0%)

Age Statistics by Gender:
        count   mean       std   min    25%   50%   75%   max
Gender
Female   30.0  66.70  9.086367  50.0  60.00  67.0  75.5  79.0
Male     20.0  67.85  8.981121  51.0  60.75  70.0  75.0  79.0

Age Group Distribution:
  50-59: 11 (22.0%)
  60-69: 14 (28.0%)
  70-79: 25 (50.0%)

✓ Demographic visualization saved: 'demographic_analysis.png'

======================================================================
2. HEALTH METRICS OVERVIEW
======================================================================

Health Metrics Summary Statistics:
       DailyStepCount  HeartRate_Avg  ...  CalorieBurn  WeeklyHealthScore
count       50.000000        50.0000  ...    50.000000           50.00000
mean      7482.260000        82.2800  ...  2569.720000           77.88000
std       2830.756644        11.1246  ...   606.615503           13.12054
min       2064.000000        65.0000  ...  1511.000000           56.00000
25%       5100.250000        71.0000  ...  2107.000000           68.00000
50%       7783.500000        83.0000  ...  2543.500000           77.50000
75%       9910.000000        92.0000  ...  3118.750000           86.00000
max      11762.000000        99.0000  ...  3452.000000           99.00000

[8 rows x 7 columns]

Activity Level Distribution:
  Sedentary: 12 (24.0%)
  Moderate: 26 (52.0%)
  Active: 12 (24.0%)

Sleep Category Distribution:
  Good: 23 (46.0%)
  Poor: 14 (28.0%)
  Fair: 13 (26.0%)

Stress Level Distribution:
  Low: 12 (24.0%)
  Moderate: 17 (34.0%)
  High: 21 (42.0%)

Health Score Categories:
  Poor: 4 (8.0%)
  Fair: 20 (40.0%)
  Good: 15 (30.0%)
  Excellent: 11 (22.0%)

✓ Health metrics visualization saved: 'health_metrics_distribution.png'

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======================================================================
3. CORRELATION ANALYSIS
======================================================================

Top 10 Strongest Correlations (excluding diagonal):
  DailyStepCount <-> Risk_Score: r = -0.672
  StressIndex <-> Risk_Score: r = 0.569
  WeeklyHealthScore <-> Risk_Score: r = -0.463
  HeartRate_Max <-> Risk_Score: r = 0.315
  HeartRate_Min <-> StressIndex: r = 0.314
  CalorieBurn <-> Risk_Score: r = -0.298
  DailyStepCount <-> HeartRate_Max: r = -0.292
  SleepDuration_Hours <-> Risk_Score: r = -0.290
  HeartRate_Min <-> Risk_Score: r = 0.285
  HeartRate_Max <-> WeeklyHealthScore: r = -0.284

Key Correlation Insights:
  • Sleep Duration & Stress Index: r = -0.230
    Moderate inverse relationship
  • Daily Steps & Health Score: r = -0.021
    Negative association
  • Sleep Quality & Health Score: r = 0.181

✓ Correlation heatmap saved: 'correlation_heatmap.png'

======================================================================
4. AGE GROUP TRENDS
======================================================================

Average Health Metrics by Age Group:
          DailyStepCount  HeartRate_Avg  ...  CalorieBurn  WeeklyHealthScore
AgeGroup                                 ...
50-59            7397.18          83.00  ...      2768.64              77.82
60-69            6711.57          80.21  ...      2562.79              78.71
70-79            7951.28          83.12  ...      2486.08              77.44

[3 rows x 7 columns]

ANOVA Tests for Age Group Differences:
  WeeklyHealthScore: F=nan, p=nan ns
  StressIndex: F=nan, p=nan ns
  SleepDuration_Hours: F=nan, p=nan ns
  DailyStepCount: F=nan, p=nan ns

✓ Age group trends saved: 'age_group_trends.png'

======================================================================
5. GENDER COMPARISON
======================================================================

Average Health Metrics by Gender:
        DailyStepCount  HeartRate_Avg  ...  CalorieBurn  WeeklyHealthScore
Gender                                 ...
Female          6942.9          82.83  ...      2559.93               79.4
Male            8291.3          81.45  ...      2584.40               75.6

[2 rows x 7 columns]

Independent T-Tests for Gender Differences:
  DailyStepCount: t=1.681, p=0.0994 ns
  HeartRate_Avg: t=-0.427, p=0.6712 ns
  SleepDuration_Hours: t=-0.804, p=0.4256 ns
  SleepQualityScore: t=-0.100, p=0.9211 ns
  StressIndex: t=0.145, p=0.8854 ns
  CalorieBurn: t=0.138, p=0.8906 ns
  WeeklyHealthScore: t=-1.003, p=0.3207 ns

✓ Gender comparison saved: 'gender_comparison.png'
```

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======================================================================
6. FALL ALERT RISK FACTORS
======================================================================

Patients with Fall Alerts: 5 (10.0%)

Fall Alert Patients Characteristics:
  Average Age: 60.8 vs 67.9 (no falls)
  Average Steps: 9236 vs 7287
  Average Sleep: 6.9 vs 6.9 hours
  Average Health Score: 79.4 vs 77.7

Fall Alerts by Age Group:
  50-59: 3/11 (27.3%)
  60-69: 1/14 (7.1%)
  70-79: 1/25 (4.0%)

Fall Alerts by Activity Level:
  Sedentary: 0/12 (0.0%)
  Moderate: 3/26 (11.5%)
  Active: 2/12 (16.7%)

✓ Fall alert analysis saved: 'fall_alert_analysis.png'

======================================================================
7. PATIENT RISK SEGMENTATION
======================================================================

Risk Category Distribution:
  Low Risk: 6 (12.0%)
  Medium Risk: 18 (36.0%)
  High Risk: 19 (38.0%)
  Critical Risk: 7 (14.0%)

Risk Distribution by Age Group:
RiskCategory  Critical Risk  High Risk  Low Risk  Medium Risk
AgeGroup
50-59                  18.2       27.3       0.0         54.5
60-69                   7.1       57.1       7.1         28.6
70-79                  16.0       32.0      20.0         32.0

High-Risk Patient Profile (n=26):
  Average Age: 67.3 years
  Average Steps: 6109
  Average Sleep: 6.6 hours
  Average Stress: 70.3
  Average Health Score: 73.4
  Fall Alert Rate: 11.5%

✓ Risk segmentation saved: 'risk_segmentation.png'

======================================================================
8. SLEEP-STRESS CORRELATION ANALYSIS
======================================================================

Pearson Correlation: r = -0.230
Linear Regression: StressIndex = 83.49 + -3.43 * SleepDuration
R² = 0.053, p-value = 0.1083

Average Stress by Sleep Category:
                mean    std  count
SleepCategory
Fair           68.54  14.58     13
Good           52.39  20.30     23
Poor           63.79  22.41     14

✓ Sleep-stress analysis saved: 'sleep_stress_analysis.png'

======================================================================
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9. EXPORTING SUMMARY STATISTICS
======================================================================

✓ Summary statistics exported: 'eda_summary_statistics.xlsx'

======================================================================
KEY INSIGHTS FROM EXPLORATORY ANALYSIS
======================================================================

1. DEMOGRAPHIC INSIGHTS:
   • Gender distribution: 30 Female, 20 Male
   • Age range: 50-79 years (mean: 67.2)
   • Most common age group: 70-79

2. HEALTH STATUS:
   • Average health score: 77.9/100
   • High-risk patients: 26 (52.0%)
   • Fall alert incidence: 10.0%

3. ACTIVITY PATTERNS:
   • Average daily steps: 7482
   • Sedentary patients: 12 (24.0%)
   • Active patients: 12 (24.0%)

4. SLEEP & STRESS:
   • Average sleep duration: 6.9 hours
   • Average stress index: 59.8/100
   • Sleep-stress correlation: r = -0.230 (strong inverse)

5. RISK FACTORS:
   • Sleep duration < 6h: 14 patients
   • High stress (>70): 20 patients
   • Low activity (<5000 steps): 12 patients

======================================================================
EXPLORATORY DATA ANALYSIS COMPLETE
======================================================================

Generated Outputs:
  1. demographic_analysis.png
  2. health_metrics_distribution.png
  3. correlation_heatmap.png
  4. age_group_trends.png
  5. gender_comparison.png
  6. fall_alert_analysis.png
  7. risk_segmentation.png
  8. sleep_stress_analysis.png
  9. eda_summary_statistics.xlsx

Next Steps:
  • Build predictive models
  • Perform clustering analysis
  • Create interactive dashboards
======================================================================

Process finished with exit code 0
```